

Muscle Memory Acquisition in Virtual Reality vs. Real-World Environments

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Abstract

This study compares fine motor muscle memory acquisition in a Virtual Reality environment with real-world practice. Twelve participants were split into two groups and learned a dot pattern similar to that used on a phone lock screen. One group learned the pattern in a VR environment, using a Quest 3 headset and Unity. The other group learned the pattern in the real world, using a whiteboard and a whiteboard marker. Both groups took the same paper-based test with the same pattern at the end of the practice rounds. Success was measured by accuracy and completion time. Both groups achieved 100% accuracy, and mean completion time did not differ significantly (RW: 3.97s, VR: 4.22s; $p = 0.804$). This suggests VR could be useful for learning fine motor skills. However, the study found that prior VR experience was a strong predictor of performance, accounting for 77.9% of the variance in final completion time ($F(2,9) = 15.83$, $p = 0.001$). These findings suggest that future VR motor learning studies may need to control for prior VR experience as a confounding variable.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Virtual reality*.

Keywords

Virtual Reality, muscle memory, motor skill learning, HCI, pattern drawing

ACM Reference Format:

Riley Nelson. 2026. Muscle Memory Acquisition in Virtual Reality vs. Real-World Environments. In *Proceedings of Human-Computer Interaction (CS 465 '25)*. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/nnnnnnnn>. nnnnnnn

1 Introduction

Motor skill learning is an important part of much of daily and professional life, from everyday tasks like walking and driving to high-stakes professional skills in fields such as medicine, sports, and the arts [9]. These skills are typically acquired through experience

and practice, often involving significant time, resources, and in some cases risk-taking [9].

This study aims to explore the effectiveness of Virtual Reality as a learning and training tool for fine motor acquisition. Virtual Reality is already being applied in diverse training contexts, including training military physicians, naval craft operators, and car drivers [4]. Previous studies and applications have demonstrated the possible benefits of using VR training software as a safer alternative to real-world training. VR training also reduces reliance on physical supplies as everything is done virtually [2]. By creating realistic training environments in Virtual Reality, we may be able to help people build motor skills in a unique and engaging environment.

The central question of this research is whether someone can acquire a fine motor muscle memory skill as quickly and accurately in Virtual Reality as they can by practicing that same skill in the real world. This has implications for the use of Virtual Reality in career training, education, and day-to-day life [4, 6].

2 Related Work

Virtual Reality training software is already being used in both rehabilitation and skills training. VR applications have been shown to be engaging and motivating, and in some cases similar to or potentially even better than conventional methods [3]. Previous Virtual Reality studies have demonstrated its possible effectiveness in applications ranging from complex sports movements [7] to neurorehabilitation [5]. Kapiotis and Theodorakou [4] note that VR has been applied in diverse training contexts, including the instruction of military physicians, naval craft operators, and car drivers, with early use of VR-based pilot training dating back to around 1990. Fitton speculates that the use of VR in training contexts is due to its convenience and because it can be a cost-effective alternative for high-risk or resource-intensive situations [2].

The most common approach to teaching motor skills in VR is active learning or learning through rehearsal from a first-person perspective [2]. Input tools supporting this approach include head-mounted displays, body gloves, and data gloves, all of which have been used in VR motor skill training contexts [4]. Access to such tools was limited for this study. Therefore, the use of controllers was chosen as the input device. In retrospect, the use of a pen may have been a more accurate tool as it more closely resembles a whiteboard marker. A head-mounted display is also used to provide users with a more immersive, classroom-like experience.

Watson and Livingston [10] examined the use of different mixed reality displays, including Virtual Reality, in the learning of motor skills, a study especially helpful in designing the present experiment. They identified several important confounding factors, including

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CS 465 '25, Fort Collins, CO

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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
<https://doi.org/10.1145/nnnnnnnn>

the importance of tools not interfering with the learning experience, avoiding pulling participants out of the virtual experience, and effectively modeling movements throughout the study [10]. Their findings showed no significant difference in motor skill recall between mobile VR, desktop PC, and video platforms, suggesting that VR is a comparable method for delivering observational motor learning content [10].

VR training is not without its challenges. Brazil and Rys [1] found that participants performing fine motor tasks in VR could not reach the same level of performance as in real life, and that learning the VR system itself can be mistaken for actual task improvement if not carefully accounted for. Because of this, the experience level of participants was taken into account for this study. Additionally, some participants reported cybersickness, which creates further limitations for VR-based motor training [1]. With this in mind, the prototype employs as little movement as possible to help reduce cybersickness in participants, although it may not eliminate it entirely. Research on skill transfer from VR to the real world has also shown mixed results, with some studies showing positive transfer and others showing no transfer or even negative transfer [6]. Juliano and Liew [3] found that while motor skill acquisition occurred at the same rate in both HMD-VR and conventional screen environments, skills acquired in HMD-VR did not fully transfer to a conventional screen environment. Conversely, skills acquired in a conventional screen environment not only transferred to HMD-VR but actually improved in the new environment [3]. Personal factors including presence, gender, age, and video game experience were found to predict the degree of transfer, suggesting that individual differences may act as meaningful confounding variables in studies comparing VR and real-world motor learning [3].

Perry et al. [8] employed the use of functional near-infrared spectroscopy to measure brain activity during VR dental tasks, finding that individual learning styles influenced the degree of neural engagement in working memory regions. This also shows that individual differences may affect how people acquire fine motor skills in VR, and could act as a confounding variable [8]. These confounding variables may be difficult to control for entirely, but are important to note in VR training research.

While prior research has demonstrated VR shows promise as a training environment, fewer studies directly compare motor memory acquisition in immersive VR environments to equivalent real-world practice using fine motor tasks [1, 3]. This project situates itself within the field of human-computer interaction by examining how interface design, controller input, and immersive visual feedback influence motor skill learning.

3 Method

3.1 Participants

Twelve participants participated in this study (n=6 VR, n=6 RW), ranging in age from 25 to 75 years. Participants included males, females, and non-binary individuals. VR experience varied widely across both groups, from participants who owned a VR headset and used it weekly (Lots, n=3) to participants with some prior experience (Some, n=6) to participants with no prior VR experience (None, n=3). Participants were recruited through convenience sampling and randomly assigned to either the VR or real-world group.

3.2 Task Design

This study compares how well participants learn a muscle memory skill in Virtual Reality versus the real world. To ensure the task is transferable between both environments, a pattern-drawing exercise was designed, inspired by the pattern lock software found on smartphones. Participants connect dots arranged in a grid using lines, forming a specific pattern. The task is meant to be intuitive enough that the tools used do not distract from the skill being tested, while the pattern itself can be made complex enough to challenge participants.

3.3 Virtual Reality Environment

The Virtual Reality environment was created using Unity. Participants used a Quest 3 headset in a virtual environment modeled after a classroom. Participants walked through a tutorial demonstrating how to use the controllers, including how to move around, turn, select items, hover, and connect dots.

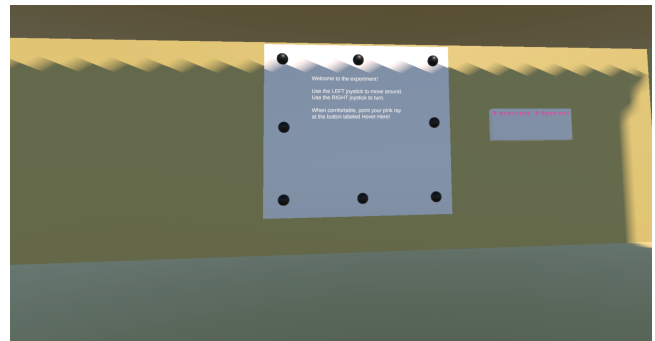


Figure 1: The VR tutorial environment where participants learned to use the Quest 3 controllers before beginning the experiment.

After the tutorial, participants pressed a button to enter the experiment. A demo video demonstrating the pattern was projected on a board at the front of the virtual room, walking participants through the experiment process and how to complete the pattern. This same video was shown to both VR and real-world participants. Participants then replicated the pattern on a virtual whiteboard next to the projected video using the VR controller.

A whiteboard view was chosen to avoid participants having to look downward in order to draw the pattern. Dots were displayed in a grid pattern on the whiteboard, and line-snapping functionality was included to make drawing simpler for those who were not familiar with VR. A three-second countdown alerted participants when to start, after which the program automatically recorded their time. Participants practiced until completing three successful attempts. Practice time was recorded as an additional data point. This was collected to see whether the amount of time practiced had an effect on final performance. After the third successful attempt, the screen displayed the amount of time spent practicing and the number of successful and unsuccessful attempts. This information was then automatically stored in a CSV file with the participant's anonymous number. After the VR portion, each participant had

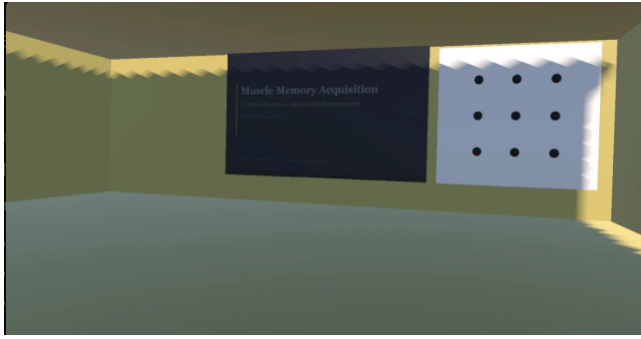


Figure 2: The VR experiment environment showing the virtual whiteboard and dot pattern that participants replicated using the controller.

one timed attempt on paper, with the researcher recording success and completion time.

3.4 Real-World Condition

The real-world participants watched the same demo video as the VR participants on a laptop screen. They were given a whiteboard with dots pre-drawn in sharpie and connected the dots using a whiteboard marker. A three-second countdown was given, after which the person conducting the study began timing using a stopwatch. Stopwatch timing introduces a small degree of human error compared to the automated timing in the VR condition, which may be a confounding factor to be aware of [1]. Participants practiced until completing three successful attempts, with the person conducting the study recording practice time, successful attempts, and unsuccessful attempts by hand. Like the VR group, real-world participants had one final timed attempt on paper at the end.

3.5 Experimental Design

Participants were split into two groups. One group learned the pattern in the real world, and the other learned the pattern using the Virtual Reality environment. The groups were split to control for pattern complexity. If both groups learned different patterns, results may reflect pattern difficulty rather than learning environment, creating a confounding variable. The independent variable was the learning environment (VR versus real-world). The dependent variables were accuracy of pattern replication and time to complete the pattern during the final timed test [1]. Participants were randomly assigned to groups to control for selection bias.

In both groups, participants began by observing the pattern in the demo video, then attempted to replicate it until succeeding three times. Practice time was recorded. At the end of the study, all participants were given a piece of paper and asked to draw the pattern from memory. This final attempt was timed and documented.

The final paper-based test was the primary measure used to record accuracy (correct or incorrect) and completion time.

Participant number: Date: Time:



☐ Correct ☐ Incorrect

Total Time: _____

Final Score: _____

Figure 3: The final paper test used to measure retention of the pattern. Participants drew the pattern from memory and were timed.

4 Results and Discussion

4.1 Accuracy

All twelve participants completed the final pattern replication task correctly (100% accuracy across both groups). Due to the lack of variance in accuracy scores, no statistical analysis was performed on this measure. This finding suggests that VR may work just as well as real-world learning environments in teaching participants to accurately learn and replicate the pattern, regardless of learning environment or VR experience level.

4.2 Final Completion Time

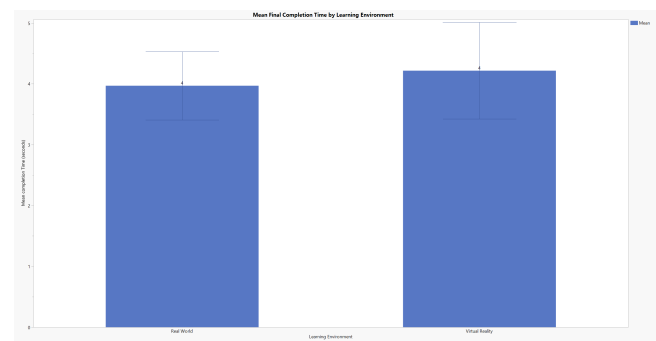


Figure 4: Mean final completion time by learning environment. Error bars show ± 1 standard error from the mean.

The mean final completion time for the real-world group was 3.97 seconds (SD = 1.43) compared to 4.22 seconds (SD = 1.74) for the VR group, a difference of 0.25 seconds or 6.3%. A pooled t-test revealed no statistically significant difference in final completion time between learning environments ($t(10) = 0.26$, $p = 0.804$, Cohen's $d = 0.147$). These results suggest that participants acquired the fine motor pattern task at a similar level regardless of whether they trained in VR or the real world. This is consistent with findings by Watson and Livingston [10] and Juliano and Liew [3] who also found VR to be a comparable method for motor skill acquisition.

4.3 Practice Time

The mean practice time for the real-world group was 19.53 seconds (SD = 7.14) compared to 22.35 seconds (SD = 7.59) for the VR group, a difference of 2.83 seconds. A pooled t-test revealed no statistically significant difference in practice time between groups ($t(10) = 0.64$, $p = 0.534$, Cohen's $d = 0.372$). However, the effect size suggests this trend may be worth examining further with a larger sample. This finding was consistent with Brazil and Rys's finding that learning the VR system itself may add to task time [1].

Three VR participants required one additional unsuccessful practice attempt compared to zero unsuccessful attempts among all real-world participants, though this difference was not formally tested due to the limited variance in the data.

4.4 Effect of VR Experience

A one-way ANOVA revealed a statistically significant effect of VR experience on final completion time ($F(2,9) = 15.83$, $p = 0.001$, $R^2 = 0.779$). VR experience accounted for 77.9% of the variance in final completion time. Post-hoc comparisons using the Tukey-Kramer HSD test showed that participants with no VR experience ($M = 6.44s$) took significantly longer than participants with some experience ($M = 3.42s$, $p = 0.002$) and participants with lots of experience ($M = 3.09s$, $p = 0.002$). Participants with some and lots of experience did not differ significantly from each other ($p = 0.842$).

VR experience also significantly predicted practice time ($F(2,9) = 5.72$, $p = 0.025$, $R^2 = 0.560$). Tukey-Kramer post-hoc comparisons revealed that participants with no VR experience ($M = 28.87s$) required significantly more practice time than participants with lots of experience ($M = 13.92s$, $p = 0.020$). Participants with some experience ($M = 20.48s$) did not differ significantly from either group. These findings align with the study conducted by Juliano and Liew [3], who found that previous HMD-VR experience predicted motor skill transfer, and with Perry et al. [8], who found that individual differences influenced performance during VR motor tasks.

4.5 Discussion

The primary finding of this study is that the learning environment alone did not significantly affect fine motor skill acquisition according to completion time or accuracy scores. Both groups achieved 100% accuracy and similar completion times, supporting the use of VR as a potential tool for learning fine motor pattern skills. This aligns with prior research demonstrating VR may be an effective training tool [3, 7, 10].

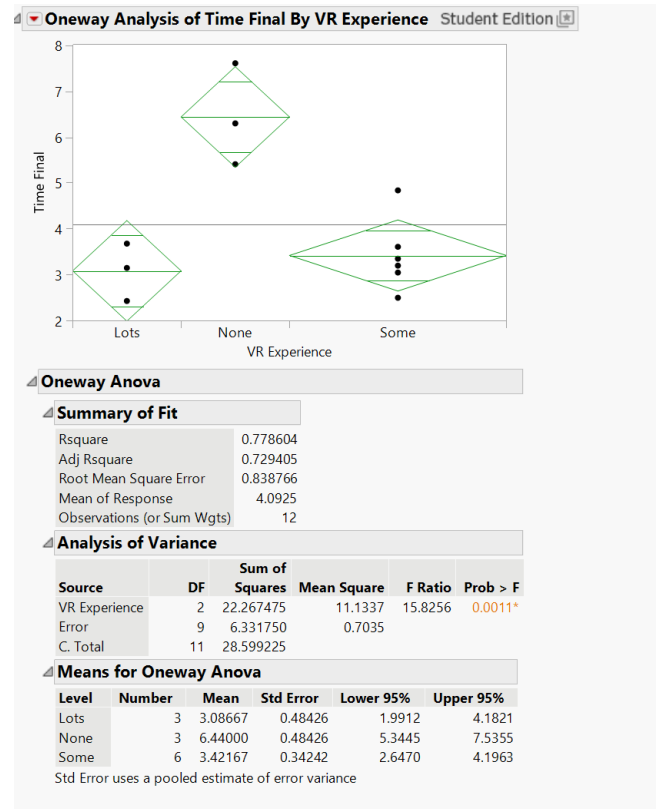


Figure 5: Oneway analysis of final completion time by VR experience level.

However, the most significant finding was the strong effect of prior VR experience on performance. Participants with no VR experience took significantly longer to complete the final task and required more practice time, regardless of which group they were assigned to. This suggests that unfamiliarity with VR interaction itself may be a barrier to performance, consistent with Brazil and Rys [1] who found that learning the VR device could be confounded with actual task performance, and with Juliano and Liew [3] who identified prior HMD-VR experience as a predictor of motor skill outcomes.

These findings highlight an important consideration for future VR motor learning studies. Prior VR experience should be controlled for or balanced across groups, as it appears to be a stronger predictor of performance than the learning environment itself. Future research with larger and more balanced samples, controlling for VR experience, would provide stronger evidence on the comparative effectiveness of VR and real-world motor skill training.

Several limitations of this study should be noted. First, the sample size was small ($n=12$, $n=6$ per group), limiting statistical power. Second, VR experience was not balanced across groups during recruitment. Third, stopwatch timing in the real-world condition introduced a small degree of human error compared to automated timing in the VR condition [1]. Fourth, the wide age range (25–75

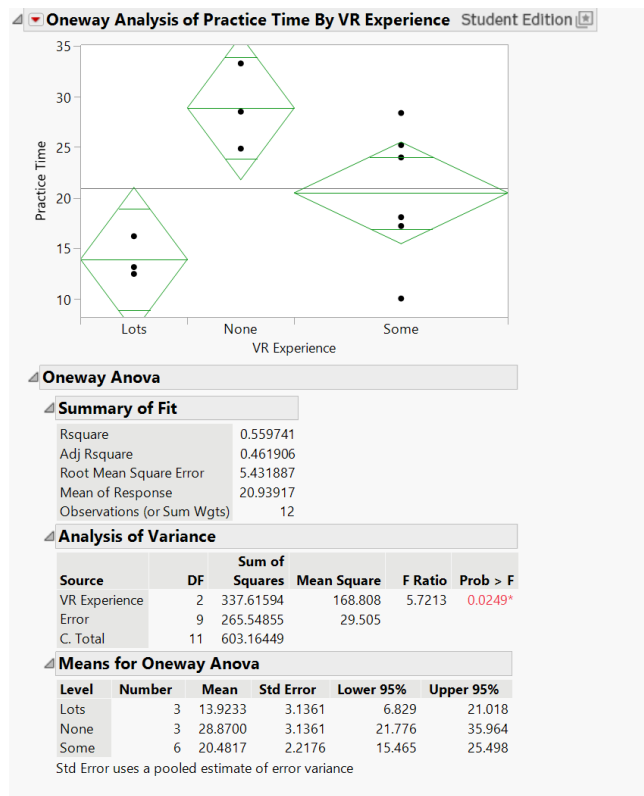


Figure 6: Oneway analysis of practice time by VR experience level.

years) and variation in VR experience represent sources of individual variability that future studies may want to control for more carefully [3]. The in-person environment was also not identical to that in the VR World, and the input devices (controllers) did not mimic that of the whiteboard marker. While this may be difficult to achieve with absolute fidelity, it is worth noting. Finally, the pattern was relatively simple, resulting in 100% accuracy across all participants. Future studies may want to use a more complex pattern to better differentiate performance between groups.

5 Conclusion

This study examined whether fine motor muscle memory skills can be acquired as effectively in a Virtual Reality environment as in the real world. Results indicated no significant difference in final completion time or accuracy between VR and real-world learning conditions, suggesting that VR may be an effective tool for fine motor skill training. However, prior VR experience proved to be a significant predictor of both final completion time and practice time, accounting for 77.9% of variance in final performance. Future VR motor learning studies should carefully control for prior VR experience as a confounding variable.

Acknowledgments

The author would like to thank the participants who contributed their time to this study, and the instructors and peers who provided feedback during the design of this experiment.

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